Abstract Algebra - Homework 2

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Problem 1

(1)

$$ij = k, \quad ij = -ji \implies -ji = k \implies ji = -k$$

Hence, $ij \neq ji$, so multiplication is not commutative and $\mathbb{H}$ is a non-commutative ring.

(2) **(a)** Let $x, y \in \mathbb{H}$. One can verify by direct computation that conjugation reverses the order of multiplication, $\overline{xy} = \bar{y}\bar{x}$. Note that for any $h = a + bi + cj + dk \in \mathbb{H}$, we have

$$N(h) = h\bar{h} = a^2 + b^2 + c^2 + d^2 \in \mathbb{R},$$

so $N(h)$ lies in the center of $\mathbb{H}$ and commutes with all elements. Using this, we compute:

$$\begin{aligned} N(xy) &= (xy)\overline{(xy)} \\ &= (xy)(\bar{y}\bar{x}) \\ &= x(y\bar{y})\bar{x} \\ &= xN(y)\bar{x} \\ &= N(y)x\bar{x} \\ &= N(y)N(x) = N(x)N(y). \end{aligned}$$

(b) Since $N(x) = x\bar{x} = a^2 + b^2 + c^2 + d^2$, it follows that

$$N(x) = 0 \iff a = b = c = d = 0 \iff x = 0.$$

Thus, $N(x) = 0$ if and only if $x = 0$.

(3) Let $x \in \mathbb{H}$ be nonzero. From previous section, we have $N(x) = x\bar{x} = a^2 + b^2 + c^2 + d^2 > 0$. Define the inverse of x by

$$x^{-1} := \frac{\bar{x}}{N(x)}.$$

Then:

$$xx^{-1} = x \cdot \frac{\bar{x}}{N(x)} = \frac{x\bar{x}}{N(x)} = \frac{N(x)}{N(x)} = 1, \quad x^{-1}x = \frac{\bar{x}}{N(x)} \cdot x = \frac{\bar{x}x}{N(x)} = \frac{N(x)}{N(x)} = 1.$$

Hence, every nonzero element of $\mathbb{H}$ is invertible.

Suppose $x, y \in \mathbb{H}$ with $x \neq 0$ and $xy = 0$. Then, $x^{-1}(xy) = (x^{-1}x)y = 1 \cdot y = y = 0$. Thus, $y = 0$, and similarly one shows that $x = 0$ if $y \neq 0$ and $xy = 0$. Therefore, $\mathbb{H}$ has no zero-divisors, and every nonzero element is invertible.

(4) *Note: The problem appears to contain a typo, We interpret* $H = \mathbb{Z} \cdot i + \mathbb{Z} \cdot j + \mathbb{Z} \cdot k + \mathbb{Z} \cdot w$.

Since each generator lies in $\mathbb{H}$, we have $H \subset \mathbb{H}$. The set H is closed under addition and subtraction because it is a $\mathbb{Z}$-module. Moreover,

$$1 = 2w - (i + j + k) \in H,$$

so H contains the identity and hence also 0.

To verify closure under multiplication, note that the products of i, j, k lie in the $\mathbb{Z}$-span of i, j, k, and known identities in the Hurwitz quaternions show that products involving w, such as w^2, iw, jw, and kw, also lie in the $\mathbb{Z}$-span of i, j, k, w. This can be verified by direct computation.

Therefore, H is a subring of $\mathbb{H}$. Since $\mathbb{H}$ has no zero-divisors and $H \subset \mathbb{H}$, it follows that H also has no zero-divisors.

(5) Fix $a, b \in H$, with $b \neq 0$. We aim to show that there exist $q, r \in H$ such that

$$a = qb + r \quad \text{with } r = 0 \text{ or } N(r) < N(b),$$

and similarly on the right: $a = bq + r$ with the same condition on r.

Since $\mathbb{H}$ is a division ring, we can write $a = q'b + r'$ for some $q' \in \mathbb{H}$, where $r' \in \mathbb{H}$. Let $q \in H$ be the Hurwitz quaternion closest to q' in the standard Euclidean norm on $\mathbb{R}^4$. Define

$$r := a - qb \in \mathbb{H}.$$

Then $N(r) = \|a - qb\|^2$ is minimized by construction, and since H is closed under subtraction, $r \in H$. Moreover, unless $q = q'$, we have $N(r) < N(b)$ because the minimal distance approximation ensures the remainder is strictly shorter than b in norm.

The same argument applies for right division by expressing $a = bq'$, choosing the nearest $q \in H$, and setting $r := a - bq$. Thus, H is both left and right Euclidean with respect to the norm N.

Problem 2

(1) We factor the polynomial:

$$f(x) = x^4 - 5x^2 + 6 = (x^2 - 2)(x^2 - 3),$$

which has roots $\pm\sqrt{2}, \pm\sqrt{3}$. These roots are not in $\mathbb{Q}$, so the splitting field is

$$E = \mathbb{Q}(\sqrt{2}, \sqrt{3}).$$

We compute the degree using the tower law:

$$\begin{aligned}
&[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = \\
&[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})] \cdot [\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = \\
&2 \cdot 2 = 4.
\end{aligned}$$

(2) Let $\alpha = \sqrt{2} + \sqrt{3}$. Then

$$\alpha^2 = (\sqrt{2} + \sqrt{3})^2 = 5 + 2\sqrt{6},$$

so

$$\sqrt{6} = \frac{\alpha^2 - 5}{2} \in \mathbb{Q}(\alpha).$$

Using $\sqrt{6} = \sqrt{2} \cdot \sqrt{3}$, we can write

$$\sqrt{2} = \frac{\sqrt{6}}{\sqrt{3}}, \quad \text{and} \quad \sqrt{3} = \alpha - \sqrt{2}.$$

Thus $\sqrt{2}, \sqrt{3} \in \mathbb{Q}(\alpha)$, so

$$\mathbb{Q}(\sqrt{2}, \sqrt{3}) \subseteq \mathbb{Q}(\alpha).$$

On the other hand, $\alpha = \sqrt{2} + \sqrt{3} \in \mathbb{Q}(\sqrt{2}, \sqrt{3})$ (since the field contains both $\sqrt{2}$ and $\sqrt{3}$, and is closed under addition), so the reverse inclusion holds. Therefore,

$$\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3}).$$